# 1.1 Proof

**What is a mathematical proof?** A proof is a logical argument that establishes the truth of a mathematical statement beyond all doubt. Unlike a demonstration or example, a proof must work for all cases covered by the statement.

## Types of Proof

| Type | Method | When to Use |
|---|---|---|
| Deduction | Start from known facts and use logical steps to reach the conclusion | General statements about all numbers/cases |
| Exhaustion | Check every possible case individually | Finite number of cases |
| Counter-example | Find one case where the statement fails | Disproving a statement |
| Contradiction | Assume the opposite is true and show this leads to an impossibility | Difficult to prove directly; irrational numbers, infinite sets |

### Choosing the Right Proof Method

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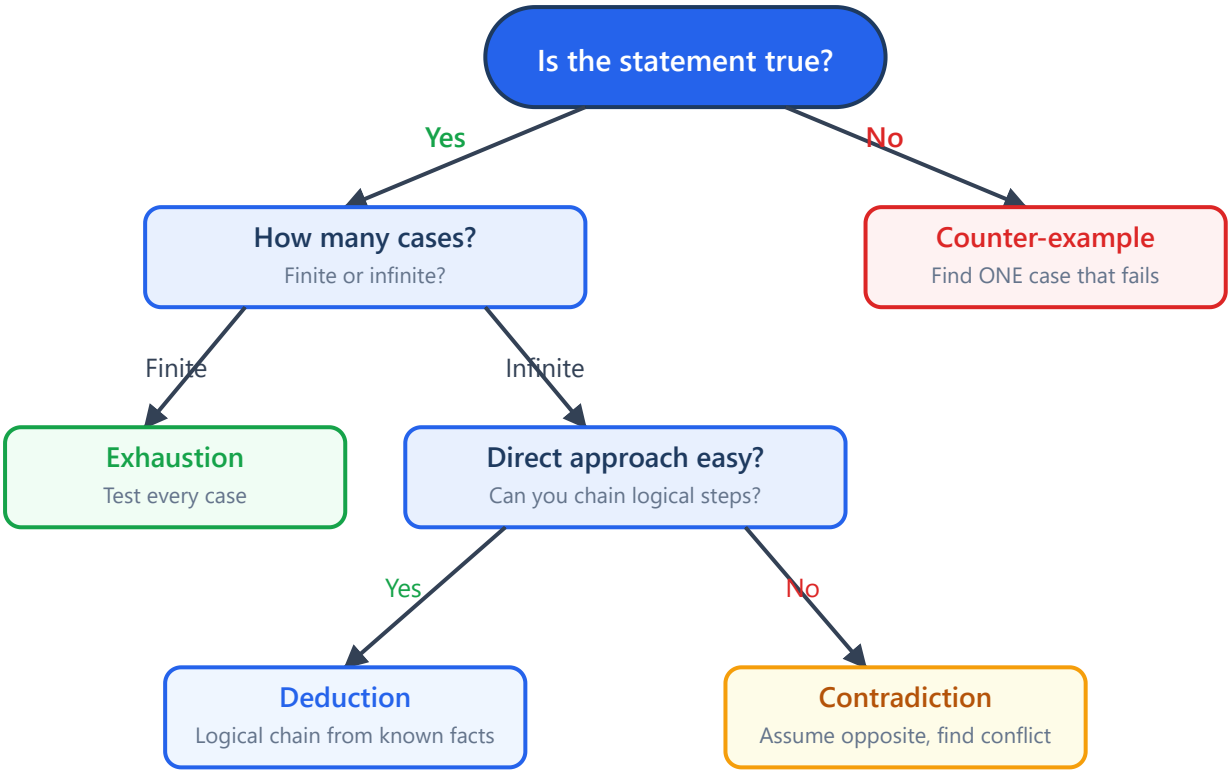



# Proof by Deduction

**Proof by deduction** starts from known mathematical facts (axioms, definitions, previously proven theorems) and uses logical steps to arrive at the required conclusion. This is the most common type of proof.

## Key Algebraic Representations

- An even number can be written as $2n$ where $n \in \mathbb{Z}$
- An odd number can be written as $2n + 1$ where $n \in \mathbb{Z}$
- Consecutive integers: $n, n + 1, n + 2, \ldots$
- A multiple of 3: $3n$ where $n \in \mathbb{Z}$
- A perfect square: $n^2$ where $n \in \mathbb{Z}$

### Example 1: Prove that the sum of two consecutive odd numbers is divisible by 4.

Let the two consecutive odd numbers be $2n+1$ and $2n+3$ where $n \in \mathbb{Z}$.

Sum = $(2n+1)+(2n+3) = 4n+4 = 4(n+1)$

Since $n+1$ is an integer, $4(n+1)$ is divisible by 4. $\square$

### Example 2: Prove that the product of two odd numbers is always odd.

Let the two odd numbers be $2a+1$ and $2b+1$ where $a, b \in \mathbb{Z}$.

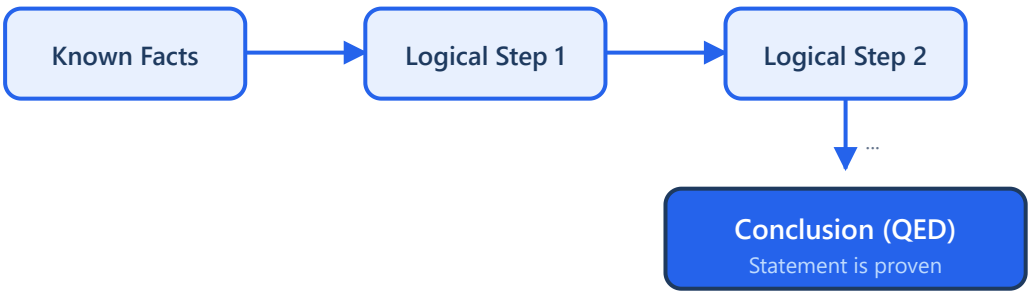
Product = $(2a+1)(2b+1) = 4ab+2a+2b+1 = 2(2ab+a+b)+1$

Since $2ab+a+b$ is an integer, this is of the form $2k+1$, which is odd. $\square$

### Example 3: Prove that $n^2+n$ is even for all $n \in \mathbb{Z}$.

Factorise: $n^2+n = n(n+1)$

This is the product of two consecutive integers. One of them must be even, so the product is even. $\square$




## Proof by Exhaustion

**Proof by exhaustion** involves checking every possible case to confirm the statement is true. This is only practical when there are a finite (and manageable) number of cases.

### Example 4: Prove that $n^2 + 2$ is not divisible by 4 for any integer $n$.

Any integer $n$ is either even or odd. We test both cases:

**Case 1:** $n$ is even, so $n = 2k$.
Then $n^2 + 2 = 4k^2 + 2 = 2(2k^2 + 1)$.
Since $2k^2 + 1$ is odd, this equals $2 \times \text{odd}$, which is not divisible by 4.

**Case 2:** $n$ is odd, so $n = 2k + 1$.
Then $n^2 + 2 = 4k^2 + 4k + 1 + 2 = 4k^2 + 4k + 3 = 4(k^2 + k) + 3$.
This leaves remainder 3 when divided by 4, so it is not divisible by 4.

All cases checked. $\square$

### Example 5: Prove that for $n \in \{1, 2, 3, 4, 5\}$, $n^2 + n + 41$ is prime.

| $n$ | $n^2 + n + 41$ | Prime? |
|---|---|---|
| 1 | 43 | Yes |
| 2 | 47 | Yes |
| 3 | 53 | Yes |
| 4 | 61 | Yes |
| 5 | 71 | Yes |

All 5 cases produce primes. $\square$

**Exam Tip:** Make sure you clearly label each case and check every single one. Examiners look for a systematic approach, not random testing. State your cases before you begin checking them.

## Proof by Counter-Example

**Proof by counter-example** is used to *disprove* a statement. You only need to find **one** example where the statement fails. This proves that the statement is not always true.

### Example 6: Disprove: "If $n$ is prime, then $2^n - 1$ is also prime."

Try $n = 11$: $2^{11} - 1 = 2047 = 23 \times 89$

Since 2047 is not prime, the statement is false. $\square$

### Example 7: Disprove: "The sum of two irrational numbers is always irrational."

Let $a = \sqrt{2}$ and $b = -\sqrt{2}$. Both are irrational.

Sum: $a + b = \sqrt{2} + (-\sqrt{2}) = 0$, which is rational.

Counter-example found, so the statement is false. $\square$

### Example 8: Disprove: "$n^2 > n$ for all integers $n$."

Try $n = 0$: $0^2 = 0$, and $0 \not> 0$. The statement fails. $\square$

(Also fails for $n = 1$: $1^2 = 1 \not> 1$)

**Exam Tip:** When disproving, you only need ONE counter-example. State it clearly, show the calculation, and explicitly say the statement is false. Try simple values first: 0, 1, -1, 2.

# Proof by Contradiction

**Proof by contradiction** works by assuming the opposite of what you want to prove, then showing this assumption leads to a logical impossibility (a contradiction). Since mathematics cannot contain contradictions, the original assumption must be wrong, so the statement is true.

## Steps for Proof by Contradiction

1. Assume the **negation** of the statement you want to prove.
2. Use logical reasoning from this assumption.
3. Arrive at a **contradiction** (something impossible or self-contradictory).
4. Conclude that the assumption was wrong, so the original statement is true.

### Example 9: Prove that $\sqrt{2}$ is irrational.

**Assume for contradiction** that $\sqrt{2}$ is rational.

Then $\sqrt{2} = \dfrac{a}{b}$ where $a, b \in \mathbb{Z}$, $b \neq 0$, and the fraction is in its simplest form (so $a$ and $b$ are not both even).

Squaring both sides: $2 = \dfrac{a^2}{b^2}$, so $a^2 = 2b^2$.

This means $a^2$ is even, so $a$ must be even. Write $a = 2k$.

Then $(2k)^2 = 2b^2 \Rightarrow 4k^2 = 2b^2 \Rightarrow b^2 = 2k^2$.

This means $b^2$ is even, so $b$ is even.

**Contradiction:** Both $a$ and $b$ are even, but we said the fraction was in simplest form.

Therefore $\sqrt{2}$ is irrational. $\square$

## Example 10: Prove that there are infinitely many prime numbers.

**Assume for contradiction** that there are finitely many primes: $p_1, p_2, \ldots, p_n$.

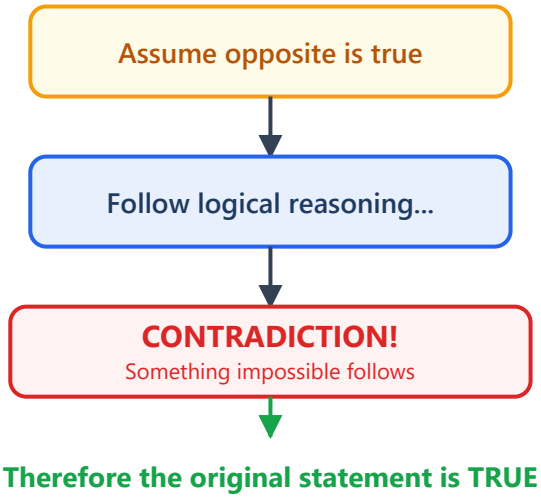
Consider the number $N = p_1 \times p_2 \times \cdots \times p_n + 1$.

$N$ is not divisible by any of $p_1, p_2, \ldots, p_n$ (since dividing always leaves remainder 1).

So either $N$ is itself prime (a new prime not in our list), or $N$ has a prime factor not in our list.

**Contradiction:** We assumed our list contained all primes.

Therefore there are infinitely many primes. $\square$




**Exam Tip:** Always begin a proof by contradiction with "Assume for contradiction that..." and end by clearly stating what the contradiction is. The examiner must see you have identified the impossibility.

# Common Proof Language

| Symbol / Phrase | Meaning |
|---|---|
| $\therefore$ | Therefore |
| $\because$ | Because |
| $\square$ or QED | End of proof ("which was to be demonstrated") |
| $\Rightarrow$ | Implies |
| $\Leftrightarrow$ | If and only if |
| $\forall$ | For all |
| $\exists$ | There exists |
| $n \in \mathbb{Z}$ | $n$ is an integer |
| WLOG | Without loss of generality |

# Proof by Induction [Year 2 — A-Level only]

**Proof by mathematical induction** proves a statement is true for all positive integers $n$ (or all integers $n \geq n_0$) by two steps:

1. **Base case:** Show the statement is true for the smallest value (usually $n = 1$).
2. **Inductive step:** Assume the statement is true for $n = k$ (the *inductive hypothesis*). Then prove it must also be true for $n = k + 1$.

Since the base case works, and each step leads to the next, the statement holds for all integers $n \geq 1$.

**Induction is like toppling dominoes**

## Standard Result: Sum of Integers

### Example 11: Prove by induction that $\sum_{r=1}^{n} r = \frac{n(n+1)}{2}$ for all $n \in \mathbb{Z}^+$.

**Base case ($n = 1$):** LHS $= 1$. RHS $= \frac{1 \times 2}{2} = 1$. LHS = RHS ✓

**Inductive hypothesis:** Assume true for $n = k$, i.e. assume $\sum_{r=1}^{k} r = \frac{k(k+1)}{2}$.

**Inductive step:** We need to show the result holds for $n = k + 1$, i.e. that $\sum_{r=1}^{k+1} r = \frac{(k+1)(k+2)}{2}$

.

$$\sum_{r=1}^{k+1} r = \sum_{r=1}^{k} r + (k+1) = \frac{k(k+1)}{2} + (k+1) = (k+1)\left(\frac{k}{2} + 1\right) = \frac{(k+1)(k+2)}{2}$$

This is the required form with $n = k + 1$. ✓

**Conclusion:** The base case holds for $n = 1$. The inductive step shows that if the result holds for $n = k$, it holds for $n = k + 1$. By the principle of mathematical induction, the result holds for all $n \in \mathbb{Z}^+$. $\square$

## Induction for Divisibility

### Example 12: Prove by induction that $6^n - 1$ is divisible by 5 for all $n \in \mathbb{Z}^+$.

**Base case ($n = 1$):** $6^1 - 1 = 5 = 5 \times 1$. Divisible by 5. ✓

**Inductive hypothesis:** Assume $6^k - 1$ is divisible by 5, i.e. $6^k - 1 = 5m$ for some integer $m$.

**Inductive step:** Consider $n = k + 1$: \[6^{k+1} - 1 = 6 \cdot 6^k - 1 = 6(6^k - 1) + 6 - 1 = 6(5m) + 5 = 5(6m + 1)\] Since $6m + 1 \in \mathbb{Z}$, this is divisible by 5. ✓

**Conclusion:** By induction, $6^n - 1$ is divisible by 5 for all $n \in \mathbb{Z}^+$. $\square$

## Induction for Matrices

**Example 13: Prove by induction that $\begin{pmatrix} 3 & 1 \\ 0 & 1 \end{pmatrix}^n = \begin{pmatrix} 3^n & \frac{3^n-1}{2} \\ 0 & 1 \end{pmatrix}$ for all $n \in \mathbb{Z}^+$**

.

**Base case ($n = 1$):** LHS = $\begin{pmatrix} 3 & 1 \\ 0 & 1 \end{pmatrix}$. RHS = $\begin{pmatrix} 3 & \frac{3-1}{2} \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 1 \\ 0 & 1 \end{pmatrix}$. ✓

**Inductive hypothesis:** Assume $\begin{pmatrix} 3 & 1 \\ 0 & 1 \end{pmatrix}^k = \begin{pmatrix} 3^k & \frac{3^k-1}{2} \\ 0 & 1 \end{pmatrix}$.

**Inductive step:** \[\begin{pmatrix}3&1\\0&1\end{pmatrix}^{k+1} = \begin{pmatrix}3^k & \frac{3^k-1}{2}\\0&1\end{pmatrix}\begin{pmatrix}3&1\\0&1\end{pmatrix} = \begin{pmatrix}3^{k+1} & 3^k + \frac{3^k-1}{2}\\0&1\end{pmatrix}\] Now: $3^k + \dfrac{3^k - 1}{2} = \dfrac{2 \cdot 3^k + 3^k - 1}{2} = \dfrac{3 \cdot 3^k - 1}{2} = \dfrac{3^{k+1} - 1}{2}$. ✓

**Conclusion:** By induction, the result holds for all $n \in \mathbb{Z}^+$. $\square$

**Proof by induction — checklist for full marks:**

- State and verify the **base case** explicitly (show LHS = RHS or state "divisible by ...").
- Write out the **inductive hypothesis** clearly: "Assume true for $n = k$, i.e. ..."
- In the inductive step, start from the *k+1* expression and use the inductive hypothesis to simplify. Do not start from what you want to prove.
- End with a clear **conclusion** invoking "by the principle of mathematical induction".

- The magic phrase for divisibility proofs: write $f(k+1) = f(k) \times (\text{something}) + (\text{extra term})$ and factor out the divisor.

## Exam Strategy Summary

**Key exam strategies for proof questions:**

- **Read the question carefully** -- "prove" means deduction/exhaustion/contradiction; "disprove" or "show by counter-example" means find a counter-example.
- **For deduction**: define variables clearly (e.g. "let $n = 2k$ where $k \in \mathbb{Z}$"), show every algebraic step, and conclude with the required form.
- **For exhaustion**: list all cases at the start, work through each systematically, tick them off.
- **For counter-example**: try small/simple numbers first. One example is sufficient.
- **For contradiction**: state your assumption clearly, work logically, and explicitly identify the contradiction.
- **For induction [Year 2]**: base case → inductive hypothesis → inductive step → conclusion. All four parts needed for full marks.
- **Always end with a concluding statement** that directly addresses the original claim.

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